

Suryanshu

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering — CGPA 8.7

Vellore, Tamil Nadu

Sep. 2022 – Ongoing

WORK EXPERIENCE

UI/UX Design Intern

PhPPoets Pvt. Ltd.

May 2024 – July 2024

Udaipur, Rajasthan

- Created **7+ custom icon packs** and **3 responsive landing pages**, delivering pixel-perfect designs for enterprise clients across multiple industries
- Developed end-to-end mobile application designs including user research, wireframing, prototyping, and **usability testing** for 2 client projects
- Designed and iterated **11+ high-fidelity prototypes** using design thinking principles to enhance user experience and improve conversion rates
- Completed **two case studies** with direct feedback from client

PROJECTS

BCP | Rust, Typescript, Python, gRPC, HTTP

Dec 2025 – Present

- Proposed and implemented BCP (Better Context Protocol), a next-generation AI agent communication protocol using **Rust**, **gRPC**, and **Protocol Buffers**, achieving **10–100x serialization performance improvements** over existing solutions (MCP/JSON-RPC); patent pending.
- Built intent-based semantic tool discovery** using embedding search and context tiering (hot/warm/cold) to dynamically reduce LLM context usage, replacing static tool-dumping approaches in AI agent pipelines.
- Architected a multi-layer security framework for an open-source AI protocol incorporating **OAuth 2.1**, mTLS, capability-based JWT tokens, and WebAssembly zero-trust sandboxing for tool execution isolation.
- Delivered a full cross-language **SDK ecosystem** (TypeScript, Python) with MCP backward-compatibility adapter, CLI tooling, VS Code extension, and **OpenTelemetry-based distributed tracing**; benchmarked end-to-end via automated test suite.

Hulaki | Go, Cobra, Fang, Bubbletea, TUI, API Testing, HTTP, GraphQL, gRPC

June 2025 – Present

- Developed high-performance terminal-native **API testing platform** using **Go** and modern CLI frameworks, delivering **3x faster** testing workflows compared to traditional GUI tools while reducing memory footprint by 80%
- Crafted intuitive **Terminal User Interface** with **Bubbletea** framework, featuring real-time syntax highlighting, interactive response visualization, and seamless keyboard navigation for enhanced developer experience
- Implemented comprehensive **multi-protocol support** spanning **RESTful APIs**, **GraphQL** queries, and **gRPC** services with advanced features including request chaining, environment variables, and automated test collections
- Created modular CLI architecture using **Cobra** command framework and **Fang** HTTP client, enabling extensible plugin system

We Go Wayback | Go, Asynq, Redis, Go-Chi, Statsd, WaybackMachine

May 2025 – June 2025

- Ported the **wayback-discover-diff** tool in **Go**, improving performance and modularity by 57%
- Built a high-performance **SimHash** calculation module for web page similarity detection, processing archived content from the Wayback Machine with optimized algorithms
- Integrated **Asynq** for managing jobs with **Redis** as backend and go routines for concurrency
- Achieved 100% portability and complete unit test coverage

TECHNICAL SKILLS

Languages: Golang, Typescript

Web Development: React, Next.js, SvelteKit, Hugo — Frontend Frameworks

Backend Technologies: Golang, Hono, Express, gRPC, REST APIs — API Development

Databases: PostgreSQL, Redis, Mongo — Database Management

DevOps & Tools: Docker, Git, GitHub Actions — CI/CD Pipeline

Cloud & Monitoring: AWS, Prometheus, Grafana — Cloud Computing

Protocols: GraphQL, WebSockets, WebRTC

UI/UX Design: Figma, Lottie — User Interface Design